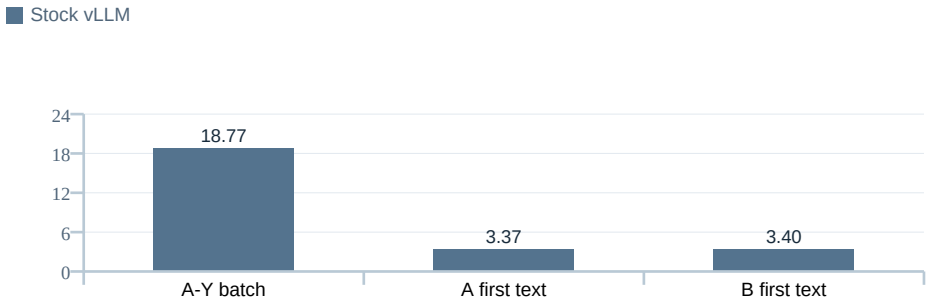
Prefix-aware request scheduling

What changed. Rank waiting requests by reusable cached prefix; break ties by FCFS. A hybrid protects the first three requests in FCFS order and applies prefix ranking to the rest, with a waiting-time safeguard.

Original concurrent A-Y workload | 25 candidates x 64 outputs | 3 running slots | 3 timing rounds per policy

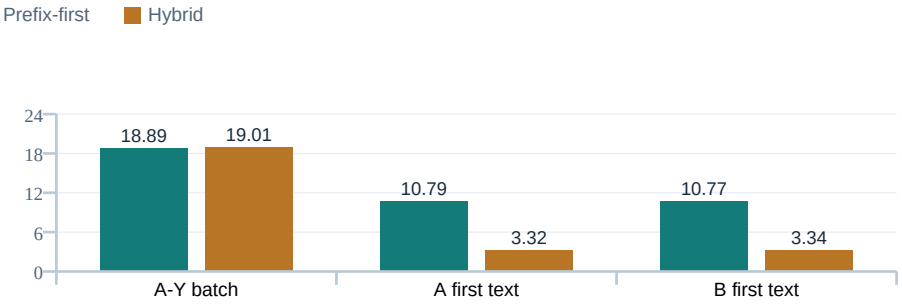
Qwen3-0.6B | Stock vLLM

seconds | lower is better



Qwen3-0.6B | Modified variants

seconds | lower is better



Qwen3-1.7B | Original / control

NOT MEASURED

No saved matched measurements for this model. Values are not estimated or borrowed from another experiment.

Qwen3-1.7B | Modified variants

NOT MEASURED

No saved matched measurements for this model. Values are not estimated or borrowed from another experiment.

Observed result. Prefix-first: batch 18.89s versus 18.77s FCFS. A/B first text rose from about 3.4s to 10.8s. Hybrid restored A/B to about 3.3s, but batch time was 19.01s.

Interpretation. No established batch-speed improvement. Early-request protection improves fairness relative to prefix-first, not overall speed. Qwen3-1.7B was not measured for this experiment.

Sources: experiments/01-prefix-scheduling/RESULTS.txt

MacBook Air M5 / Metal | three timed rounds unless stated | graph scales match within each model row; pages use different workloads

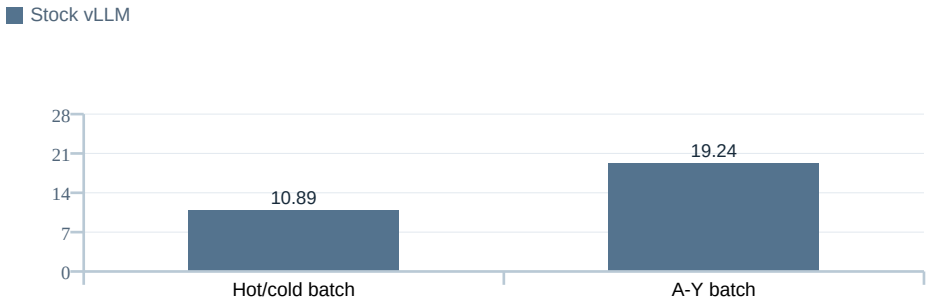
Frequency-based cache retention

What changed. Prefer evicting less frequently reused, zero-reference cached blocks. Keep actively referenced blocks ineligible for eviction and preserve FCFS request order.

Two profiles: 42 sequential hot/cold requests and original concurrent A-Y | 3 timing rounds per policy/profile

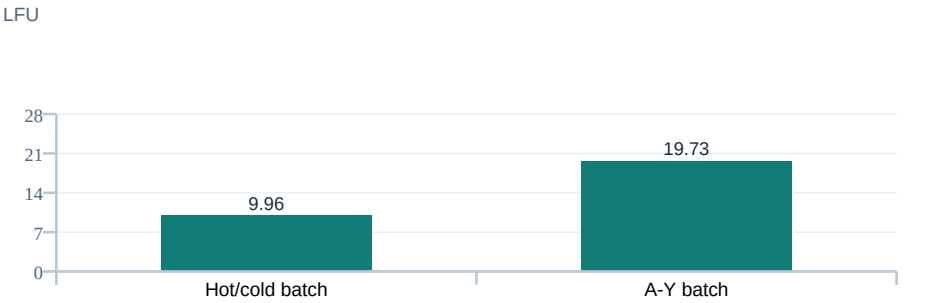
Qwen3-0.6B | Stock vLLM

seconds | lower is better



Qwen3-0.6B | Modified variants

seconds | lower is better



Qwen3-1.7B | Original / control

NOT MEASURED

No saved matched measurements for this model. Values are not estimated or borrowed from another experiment.

Qwen3-1.7B | Modified variants

NOT MEASURED

No saved matched measurements for this model. Values are not estimated or borrowed from another experiment.

Observed result. Hot-return cache reuse increased from 16 to 272 tokens; probe first text fell from 119.50ms to 52.54ms. On concurrent A-Y, reuse stayed at 2,432 tokens and LFU was 2.5% slower.

Interpretation. The hot/cold batch median fell 8.6%, but a no-reuse control also varied. Retention is demonstrated; the full timing difference is not isolated to eviction. Only 0.6B was tested.

Sources: experiments/02-frequency-eviction/RESULTS.txt | experiments/02-frequency-eviction/RESULTS-AY.txt

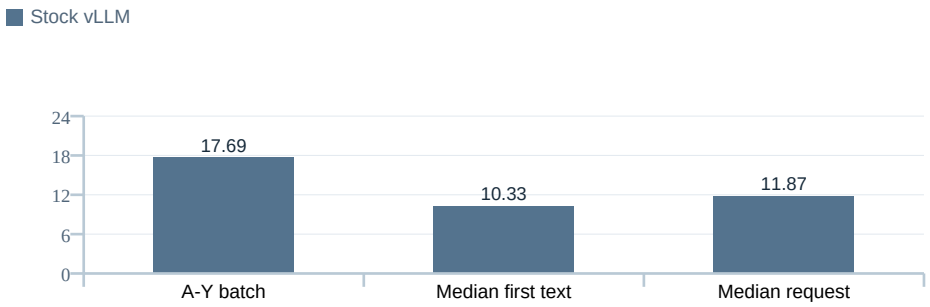
Middle-block reuse with partial repair

What changed. Match token pages even when earlier context differs, reuse their KV approximately, and recompute a boundary subset. The Metal worker skips selected rows using an alternate attention path; this is not exact causal-prefix caching.

Original A-Y | 25 candidates + 3 blockers | 96 KV pages | 3 slots | 3 matched timing pairs

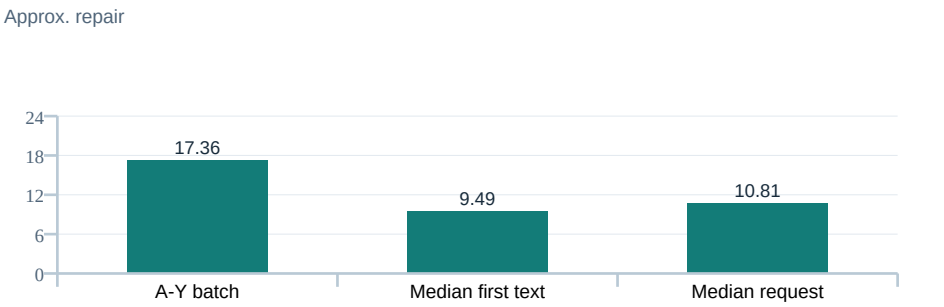
Qwen3-0.6B | Stock vLLM

seconds | lower is better



Qwen3-0.6B | Modified variants

seconds | lower is better



Qwen3-1.7B | Original / control

NOT MEASURED

No saved matched measurements for this model. Values are not estimated or borrowed from another experiment.

Qwen3-1.7B | Modified variants

NOT MEASURED

No saved matched measurements for this model. Values are not estimated or borrowed from another experiment.

Observed result. Reused 352 matching non-prefix tokens and skipped 308 rows/layer/run. Group-median batch time was 1.85% lower, but repair was slower in two of three paired rounds.

Interpretation. No reliable speedup or quality preservation established. Only 10/25 candidate answers matched stock exactly; semantic quality was not scored. No 1.7B serving comparison exists.

Sources: experiments/03-nonprefix-repair/RESULTS-VLLM-AY.txt

MacBook Air M5 / Metal | three timed rounds unless stated | graph scales match within each model row; pages use different workloads

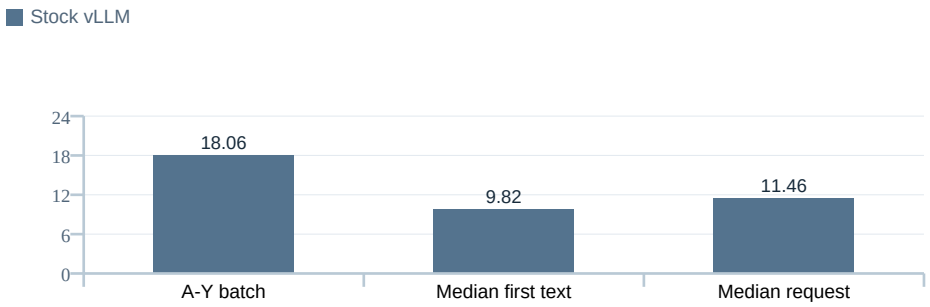
Exact partial-prefix caching

What changed. Index matching prefixes inside a 16-token physical page. Reuse the exact causal prefix and copy the partial page before continuing, preserving page ownership and invalidation. Physical pages remain 16 tokens.

Original A-Y | same native Metal worker | baseline, fine-hash control and partial reuse | 3 rotated rounds

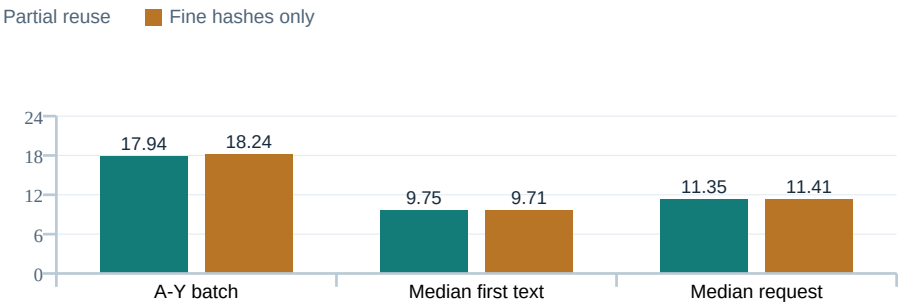
Qwen3-0.6B | Stock vLLM

seconds | lower is better



Qwen3-0.6B | Modified variants

seconds | lower is better



Qwen3-1.7B | Original / control

NOT MEASURED

No saved matched measurements for this model. Values are not estimated or borrowed from another experiment.

Qwen3-1.7B | Modified variants

NOT MEASURED

No saved matched measurements for this model. Values are not estimated or borrowed from another experiment.

Observed result. Candidate cached tokens rose from 2,432 to 2,563; the 32-token probe reused 31 instead of 16. Batch median fell 0.64%, with small improvements in all three baseline pairs.

Interpretation. The timing gain is small and exploratory. Partial reuse matched 16/25 candidate answers; fine hashes alone matched 25/25. Exact cache identity does not guarantee bitwise generation equality.

Sources: experiments/04-partial-prefix/RESULTS.txt

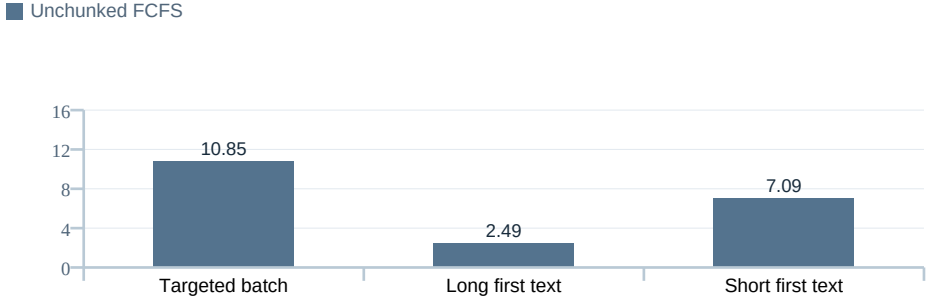
Bounded skip-and-continue scheduling

What changed. With chunked prefill disabled, bypass a waiting head that cannot fit the remaining token budget and admit a later fitting request. Restore skipped order; bound the scan to 16 requests, four bypasses/request and 30-second age protection.

Targeted A-Y: 13 long then 12 short prompts | 8 blockers/slots | 512-token batch | 3 matched pairs

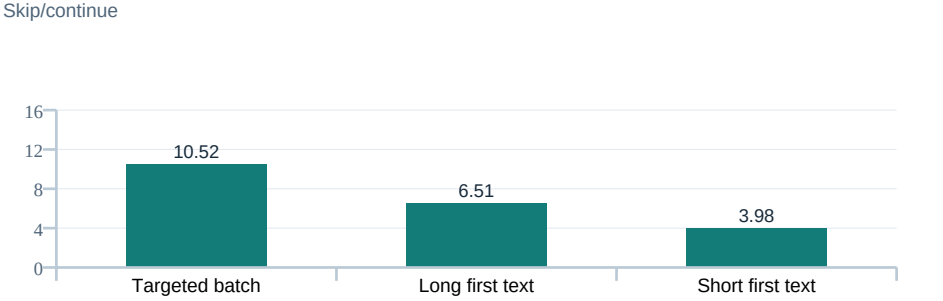
Qwen3-0.6B | Unchunked FCFS

seconds | lower is better



Qwen3-0.6B | Modified variants

seconds | lower is better



Qwen3-1.7B | Original / control

NOT MEASURED

No saved matched measurements for this model. Values are not estimated or borrowed from another experiment.

Qwen3-1.7B | Modified variants

NOT MEASURED

No saved matched measurements for this model. Values are not estimated or borrowed from another experiment.

Observed result. Batch median fell 3.04% and throughput rose 3.14%. Short-prompt first text fell 43.9%, while long-prompt first text rose 161.5%. All three targeted batch pairs were faster.

Interpretation. Original A-Y did not activate skipping. The targeted packing workload demonstrates a fairness tradeoff; the bypass bounds do not guarantee an admission deadline. Only 0.6B was tested.

Sources: experiments/05-skip-continue/RESULTS.txt

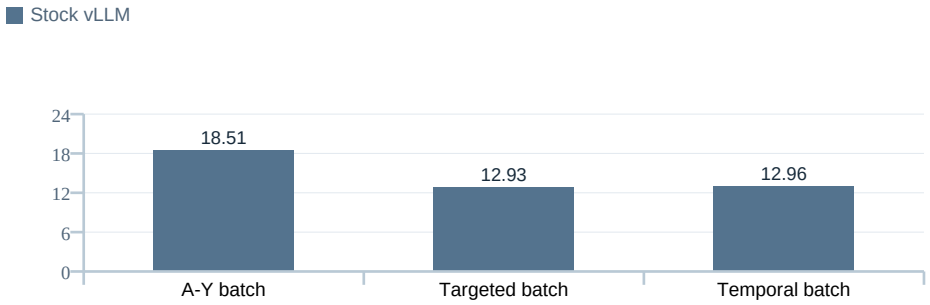
Combined cache and scheduling policies

What changed. Enable experiment 02 frequency retention, experiment 04 exact partial-prefix reuse, and experiment 05 bounded skip/continue together. Keep approximate middle-block repair disabled; both controls use unchunked prefill.

A-Y, targeted packing and sequential temporal profiles | 24 fresh-server runs including diagnostics | 3 pairs/profile

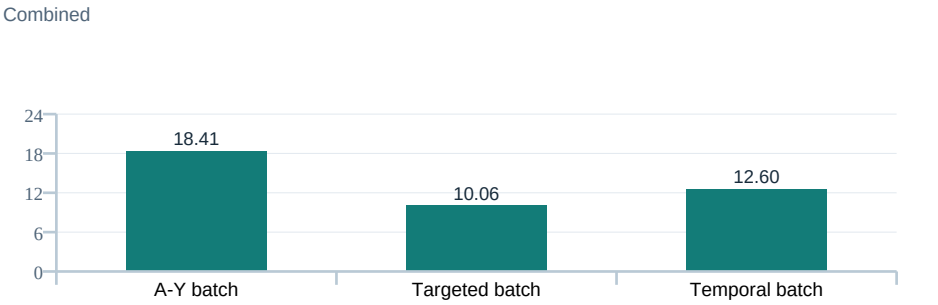
Qwen3-0.6B | Stock vLLM

seconds | lower is better



Qwen3-0.6B | Modified variants

seconds | lower is better



Qwen3-1.7B | Original / control

NOT MEASURED

No saved matched measurements for this model. Values are not estimated or borrowed from another experiment.

Qwen3-1.7B | Modified variants

NOT MEASURED

No saved matched measurements for this model. Values are not estimated or borrowed from another experiment.

Observed result. Cache gains activated consistently: A-Y +131 cached tokens, targeted +149, and hot-return reuse 16 to 277. Group-median batch times improved, but paired completion changes swung widely.

Interpretation. No stable overall speedup established. Targeted long-prompt first text rose 2.580s to 6.194s; short prompts improved. The later two-model scaling study is experiment 07, not a replication of these exact profiles.

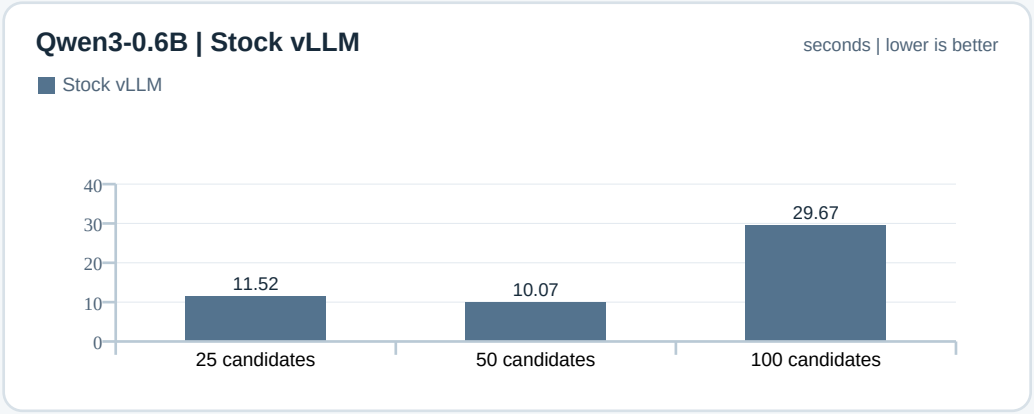
Sources: experiments/06-combined/RESULTS.txt

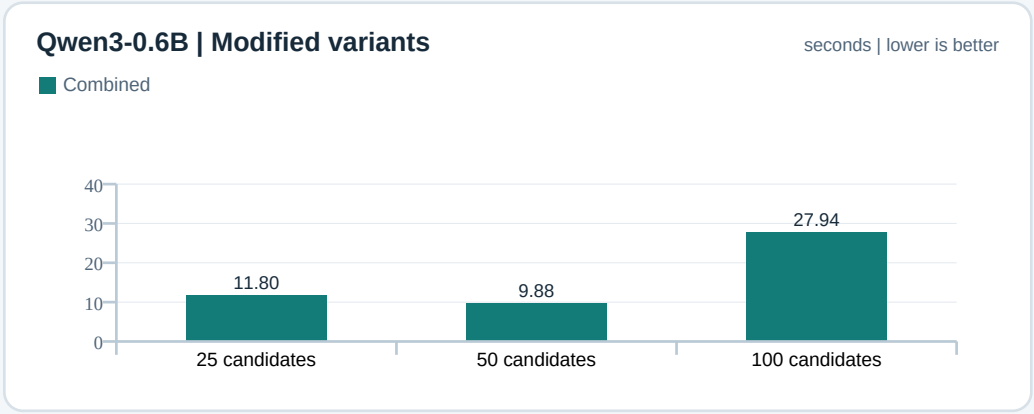
MacBook Air M5 / Metal | three timed rounds unless stated | graph scales match within each model row; pages use different workloads

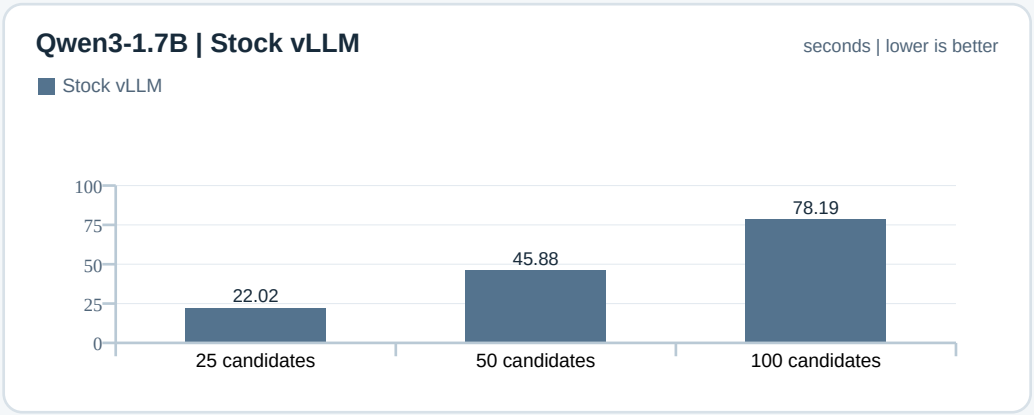
Request-count and model-size scaling

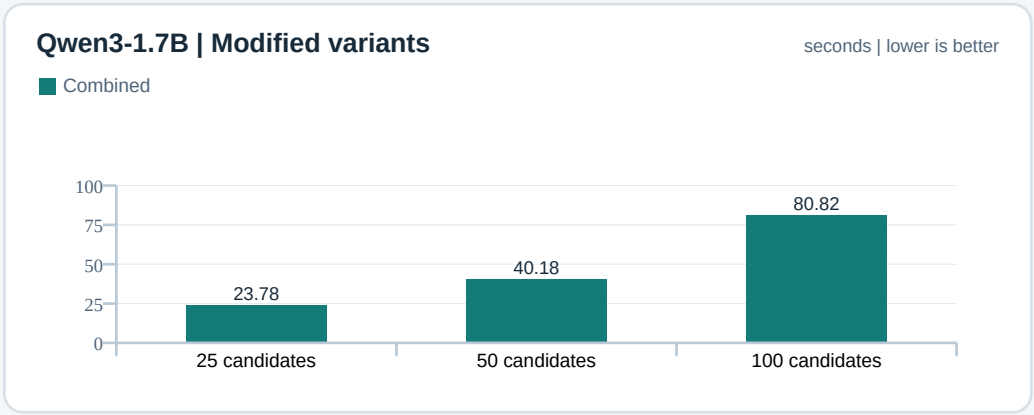
What changed. Scale the combined policy from experiment 06 to 25, 50 and 100 candidates on Qwen3-0.6B and Qwen3-1.7B. This changes workload scale and model size; it introduces no new policy.

Both pinned models | 320/64-token long/short prompts | 8 slots | 256 pages | 48 runs | 3 pairs/scale point









Observed result. Combined was faster in only 8/18 matched pairs. Cached-token gains repeated on both models: +212, +441 and +885 at 25/50/100 candidates. The 1.7B/50 group median improved 12.43%, but 1.7B/100 regressed 3.37%.

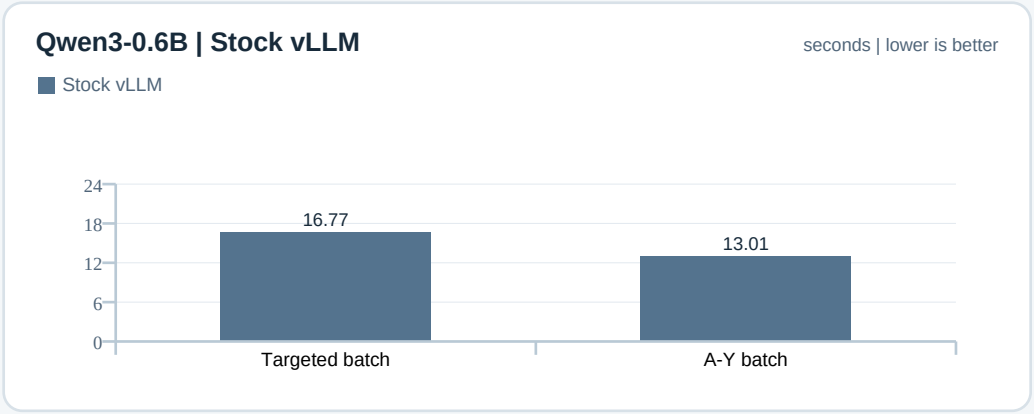
Interpretation. Run variability is large: 0.6B/50 was faster in absolute time than 0.6B/25. These bars are observed medians, not a clean scaling law or proof of a generally faster scheduler.

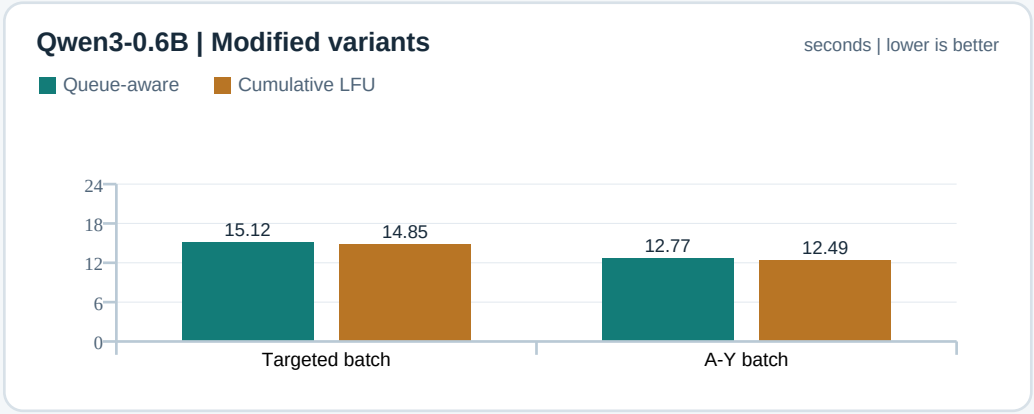
Sources: experiments/07-scale-and-model/RESULTS.txt

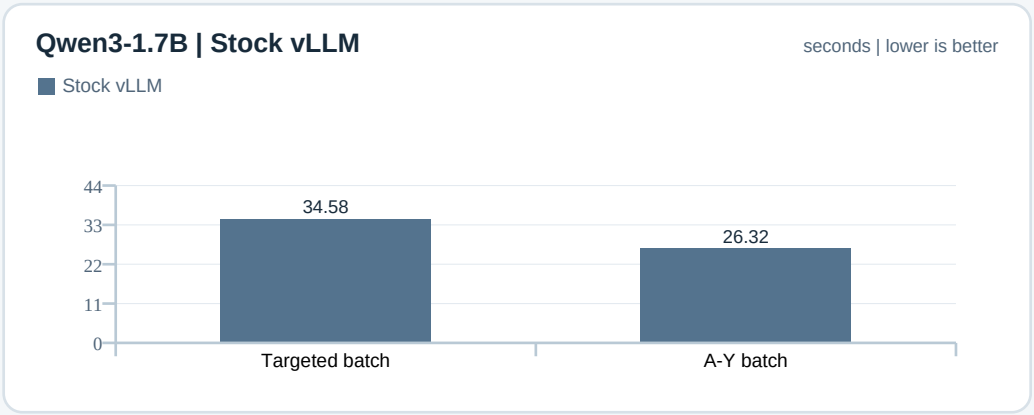
Retain cache that waiting requests need

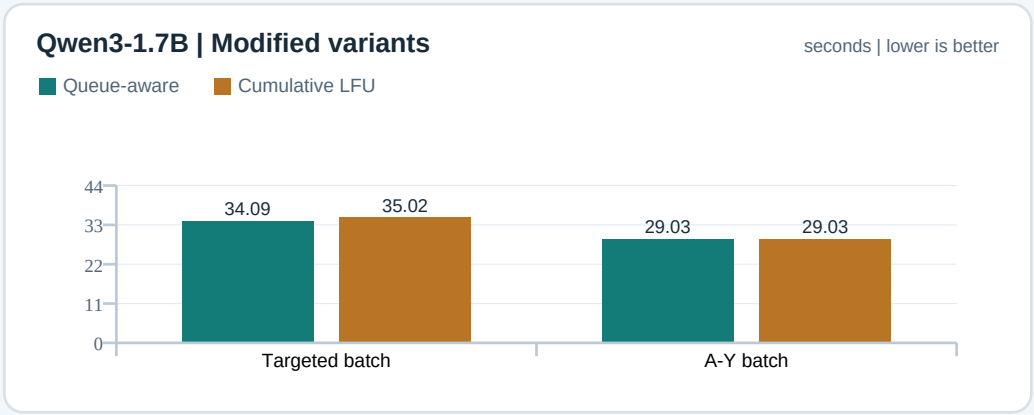
What changed. Use prefixes of already waiting requests to prefer retaining relevant zero-reference cached blocks. Keep FCFS request order and allow normal reclamation rather than permanently reserving pages.

Targeted return-prefix and original A-Y profiles | 3 policies | 24 runs/model | 3 rounds/policy/profile









Observed result. First-return reuse increased from 80 stock tokens (16 under LFU) to 176 queue-aware tokens on both models. Targeted batch median fell 9.82% on 0.6B and 1.42% on 1.7B.

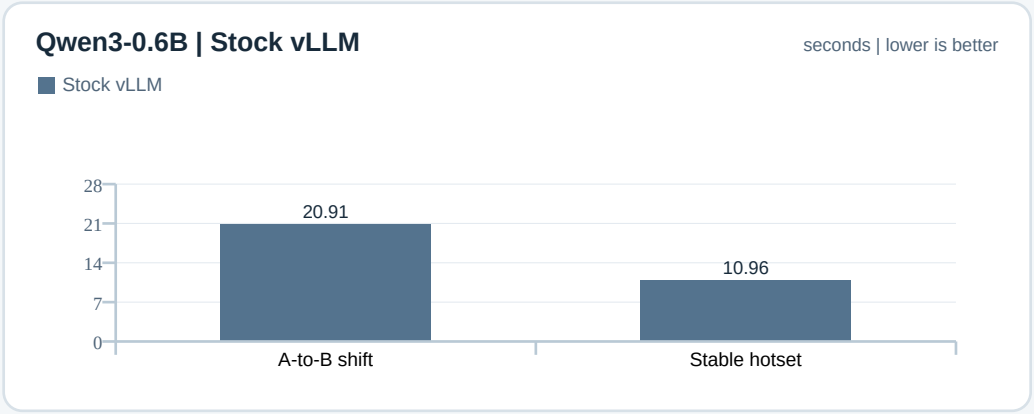
Interpretation. On A-Y, every policy reused the same 2,432 tokens. Queue-aware 1.7B A-Y was 10.30% slower and lost all three stock pairs. Retention generalizes; a consistent speed advantage does not.

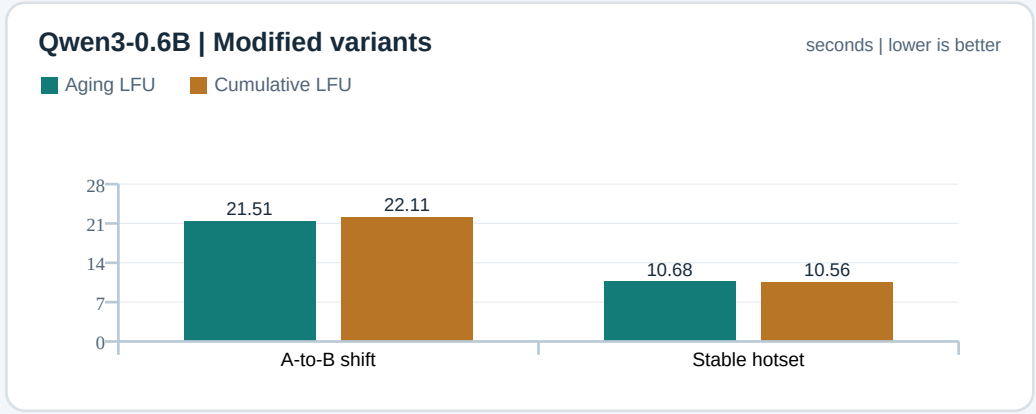
Sources: experiments/08-queue-retention/RESULTS.txt | experiments/08-queue-retention/qwen3-1.7B/RESULTS.txt

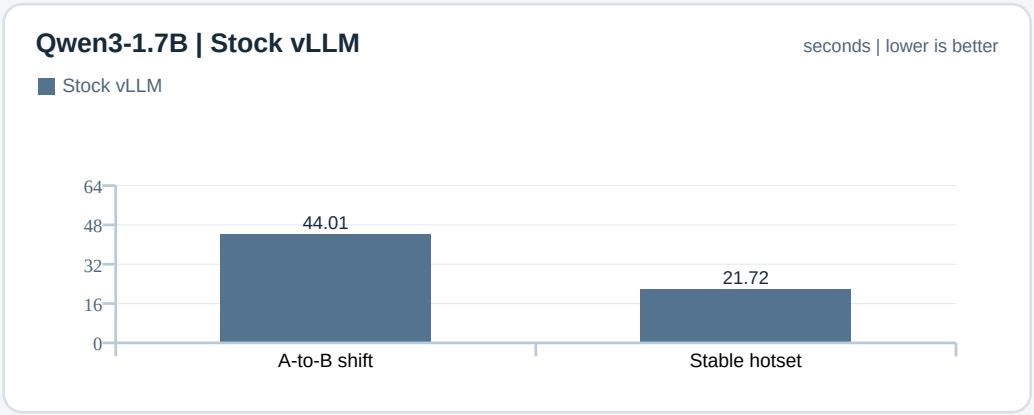
Let old cache popularity fade

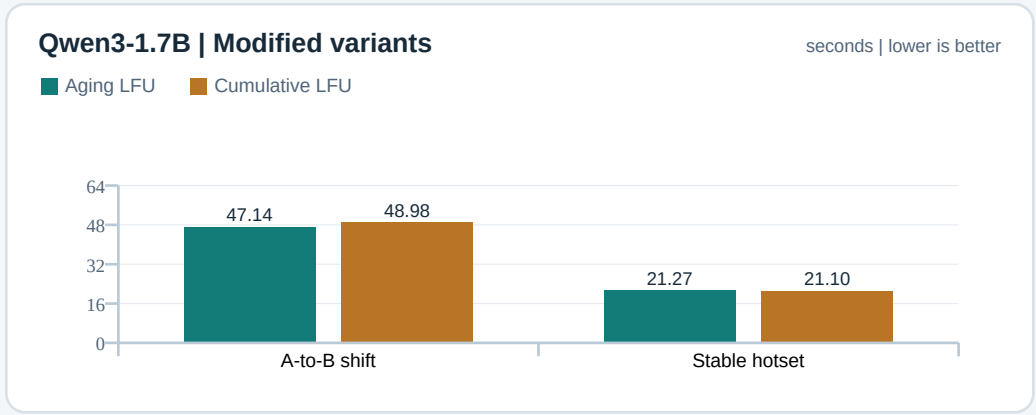
What changed. Halve all reuse scores every 64 physical-block allocations, including scores of active blocks. Prefer low-score zero-reference victims and preserve upstream free-queue tie order. Compare against stock and cumulative LFU.

84-request A-to-B shift and 42-request stable control | 32 pages | 1 slot | 24 runs/model | 3 rounds









Observed result. After cold scans, aging retained all 240 B-prefix tokens from probe cycle 2 on both models; stock retained 16 and cumulative LFU 32. Shift-probe first text improved 38.7% on 0.6B and 61.6% on 1.7B.

Interpretation. Shift batch group medians were slower than stock by 2.86%/7.10%, despite faster return probes. Stable retention stayed at 272 tokens for both LFU policies. All timed texts matched stock.

Sources: experiments/09-aging-lfu/RESULTS.txt | experiments/09-aging-lfu/qwen3-1.7B/RESULTS.txt

MacBook Air M5 / Metal | three timed rounds unless stated | graph scales match within each model row; pages use different workloads

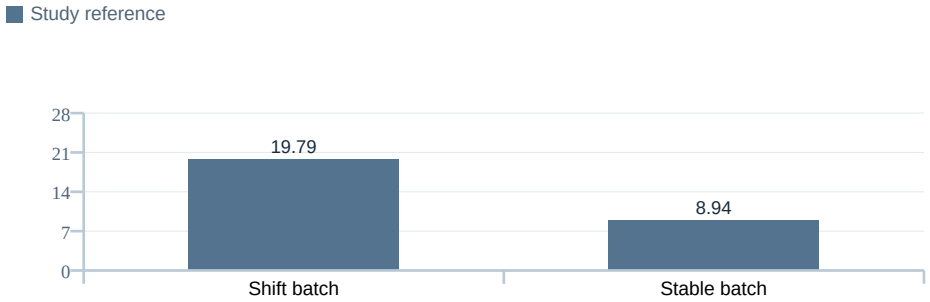
Reduce cache implementation overhead

What changed. Publish only newly computed or invalidated prompt ranges instead of rescanning prompt blocks after every step. Replace repeated free-list sorting with a lazy-invalidated LFU heap, preserving selection ties, copies and lifecycle updates.

Matched reference vs incremental LFU + partial reuse | 0.6B only | shift/stable profiles | 16 GPU runs | 3 pairs/profile

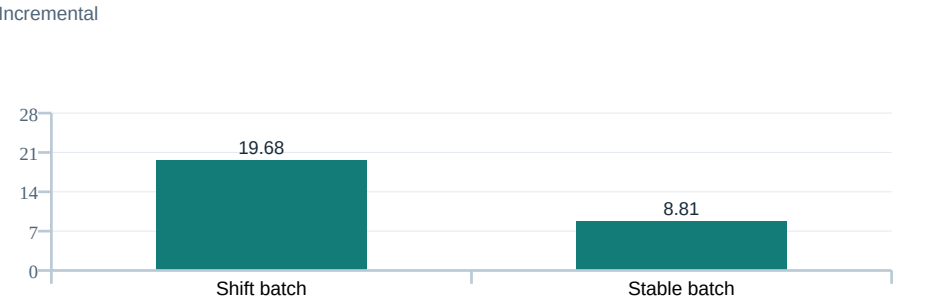
Qwen3-0.6B | Study reference

seconds | lower is better



Qwen3-0.6B | Modified variants

seconds | lower is better



Qwen3-1.7B | Original / control

NOT MEASURED

No matched reference/incremental overhead A/B. Values are not estimated or borrowed from another experiment.

Qwen3-1.7B | Modified variants

NOT MEASURED

No matched reference/incremental overhead A/B. Values are not estimated or borrowed from another experiment.

Observed result. Publisher visits fell about 87%; selector host time fell 29-37%. Batch medians improved 0.56% on shift and 1.46% on stable, but stable won only one of three matched pairs. Decisions, hits and answers matched.

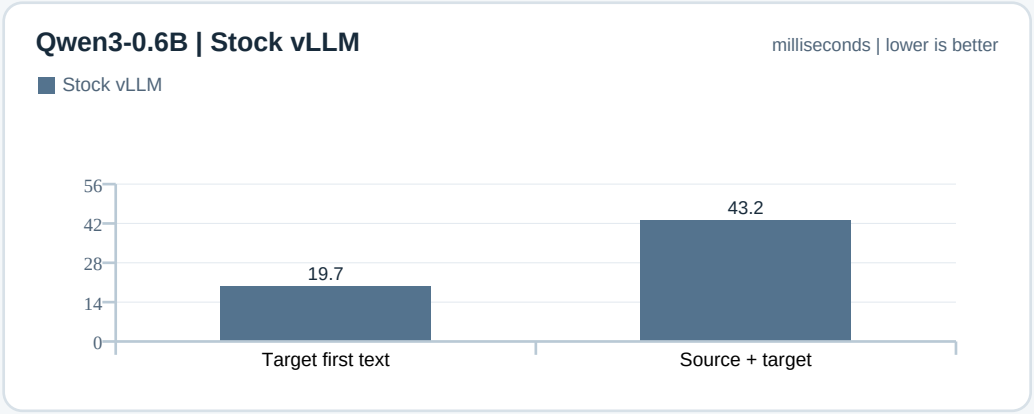
Interpretation. Control is the earlier study implementation, not stock vLLM. The 1.7B follow-up reran stock/LFU/aging, not a matched reference/incremental A/B; historical timing changes do not isolate overhead. Its shift/stable aging batches were 41.194s/22.843s vs current stock 43.729s/24.753s.

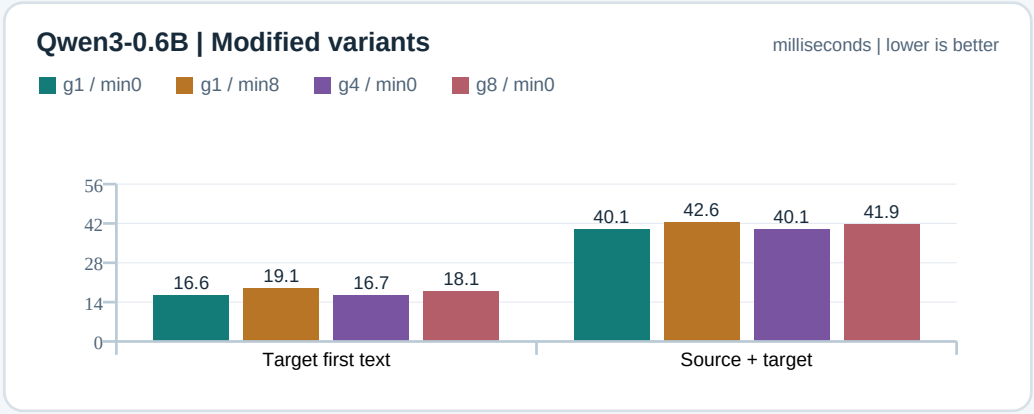
Sources: experiments/10-cache-overhead/RESULTS.txt | experiments/10-cache-overhead/aging-replication/RESULTS.txt

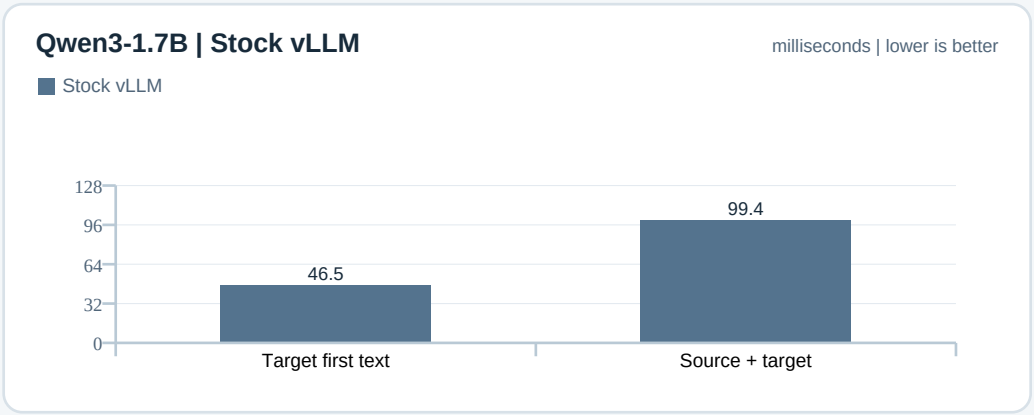
Partial-reuse thresholds and granularity

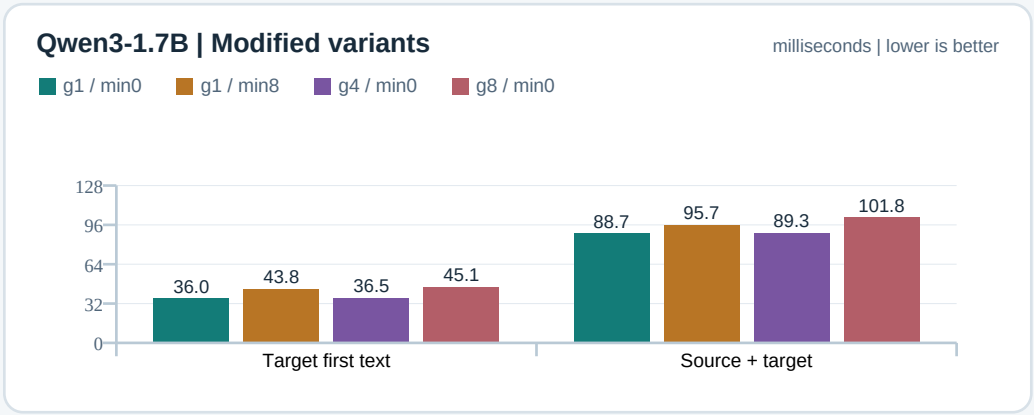
What changed. Accept partial hits only when they save a configured minimum number of extra token computations. Sweep 1/4/8-token matching and minimum 0/8/16; physical pages remain 16 tokens. min16 is a same-granularity recomputation control.

15 divergence offsets | 128-token prompts | 1 output/source and target | 135 timed targets/config | 40 runs/model









Observed result. Unrestricted g1 matching reduced target first text by 15.5% on 0.6B and 22.7% on 1.7B versus stock. Source+target time fell 7.3%/10.8%. The eight-token minimum discarded useful small hits.

Interpretation. Charts show stock and four of ten measured configurations; same-granularity min16 and other threshold controls remain in the source data. These are one-output-token probes, not concurrent throughput results. All 135 answers/config matched the recomputation control.

Sources: experiments/11-partial-threshold/RESULTS.txt | experiments/11-partial-threshold/qwen3-1.7B/RESULTS.txt

Partial reuse under concurrent mixed traffic

What changed. Carry exact one-token partial reuse into a mixed workload: unrestricted hits versus minimum eight saved tokens. Vary client concurrency while keeping four server slots and the same finite KV capacity. No new source policy is added.

1.7B only | 48 requests | 64-512 prompt / 16-64 output tokens | 16 seeded sources | 4/12 clients | 24 runs

Qwen3-0.6B | Original / control

NOT MEASURED

No saved matched measurements for this model. Values are not estimated or borrowed from another experiment.

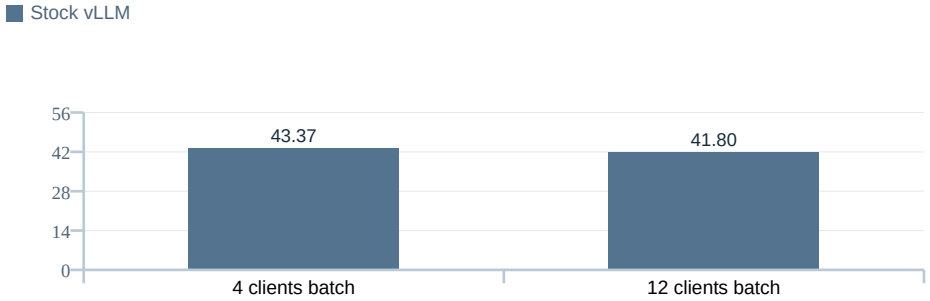
Qwen3-0.6B | Modified variants

NOT MEASURED

No saved matched measurements for this model. Values are not estimated or borrowed from another experiment.

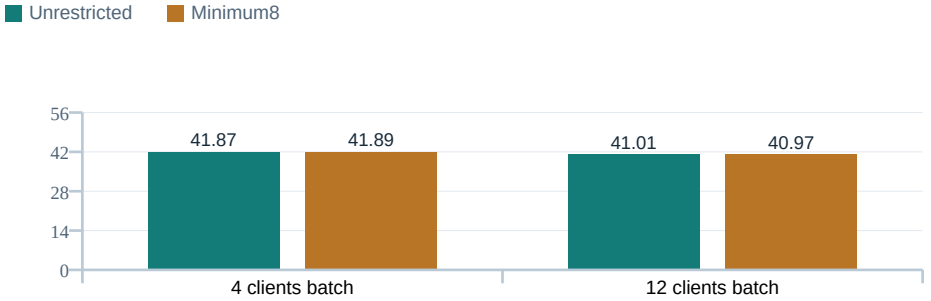
Qwen3-1.7B | Stock vLLM

seconds | lower is better



Qwen3-1.7B | Modified variants

seconds | lower is better



Observed result. Unrestricted batch medians improved 3.46% at four clients and 1.90% at twelve; minimum8 was nearly tied. At twelve clients, unrestricted p95 request latency worsened 4.81%. All paired texts matched stock.

Interpretation. Closed-loop arrivals depend on completions; the cache is seeded but churns. Seed time is excluded. Gains were not consistent in every round, and concurrency alone is not isolated from mixed lengths, longer decoding and finite-cache effects. No 0.6B replication was run.

Sources: experiments/12-concurrent-partial/RESULTS.txt

Adapt prefill chunks to current traffic

What changed. Compare stock chunked prefill with built-in fair-share adaptation and a new decoder-load cap: max(16,128/(1+2D)), clamped to the budget. An unfinished prompt older than 750ms raises the global per-request cap to at least 64.

1.7B only | 4 ongoing generators + 2 bursts of 10 long prompts | 8 slots | 128-token budget | caching/skip off | 12 runs

Qwen3-0.6B | Original / control

NOT MEASURED

No saved matched measurements for this model. Values are not estimated or borrowed from another experiment.

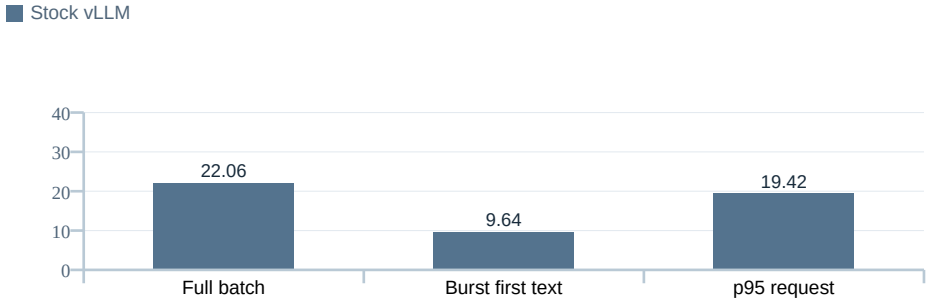
Qwen3-0.6B | Modified variants

NOT MEASURED

No saved matched measurements for this model. Values are not estimated or borrowed from another experiment.

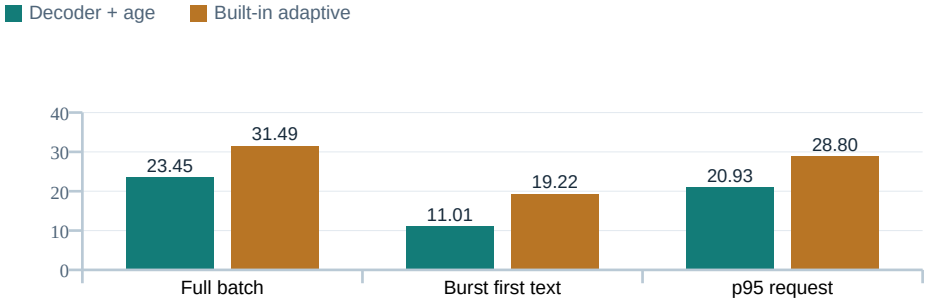
Qwen3-1.7B | Stock vLLM

seconds | lower is better



Qwen3-1.7B | Modified variants

seconds | lower is better



Observed result. The new cap reduced ongoing p95 text-chunk gaps from 201ms to 169ms (16%), but batch time rose 6.3% and burst first text 14.2%. Built-in adaptation reduced gaps to 145ms while batch time rose 42.8%.

Interpretation. This is a streaming smoothness tradeoff, not a speed win. All timed texts matched stock. Chunk gaps are not exact token ITL; the age boost is not a strict waiting bound. Three fresh-server rounds with one short warmup are exploratory.

Sources: experiments/13-adaptive-prefill/RESULTS.txt